

T3 UV one-loop RGE term comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset ig_1 Y B_\mu$
g_2	$D_\mu \supset ig_2 T^A W_\mu^A$
g_3	$D_\mu \supset ig_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$LF S_1 + \text{h.c.}$
y_2	$LF S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3Y_u^\dagger Y_u + 3Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2} \lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2} \lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2} \lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H) (S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H) (S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H) (S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H) (S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1) (S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1) (S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	crossed independent S_1 - S_2 contraction
$\lambda_{S_1}^{(A)}$	additional independent S_1 self-contraction
$\lambda_{S_2}^{(A)}$	additional independent S_2 self-contraction
α	$Y_{S_1} = \alpha/2, Y_F = (\alpha+1)/2, Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2 - 1)/4$
$T(d)$	$d(d^2 - 1)/12$
$C_{LLS_1 S_2}^{ij;AB}$	$L_i L_j S_1^A S_2^B$
C_{ijab}	$L_i L_j \phi_a \phi_b$
C_5^{ij}	$(L_i^T C L_j) H H$

Each running coupling has its own table. Rows are model configurations and columns are additive structures in the beta function. Every populated cell contains the complete model-specific term, including its numerical coefficient. A dash means that the term is absent for that configuration. All equations use $16\pi^2\mu dX/d\mu = \beta_X^{(1)}$. Gauge-coupling results use the same RGBeta $g^2 \rightarrow g$ conversion as in the previous report.

$$\beta_{g_1}^{(1)}$$

Quantity: g_1

$$16\pi^2\mu\frac{dg_1}{d\mu} = \beta_{g_1}^{(1)}$$

Column Representative term structure

T1	$\frac{55g_1^3}{6}$
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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{55g_1^3}{6}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{43g_1^3}{6}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{103g_1^3}{6}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\frac{175g_1^3}{6}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\frac{17g_1^3}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\frac{43g_1^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\frac{17g_1^3}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\frac{25g_1^3}{2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\frac{115g_1^3}{6}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\frac{21g_1^3}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{43g_1^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\frac{21g_1^3}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\frac{41g_1^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\frac{223g_1^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\frac{43g_1^3}{6}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{55g_1^3}{6}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\frac{95g_1^3}{6}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\frac{163g_1^3}{6}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{22g_1^3}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{59g_1^3}{6}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{52g_1^3}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{179g_1^3}{6}$

$\beta_{g_2}^{(1)}$
Quantity: g_2

$16\pi^2\mu\frac{dg_2}{d\mu}=\beta_{g_2}^{(1)}$

Column	Representative term structure
T1	$-\frac{g_2^3}{2}$
T2	$-\frac{3g_2^3}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{3g_2^3}{2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{g_2^3}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{g_2^3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{17g_2^3}{6}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-\frac{17g_2^3}{6}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{17g_2^3}{6}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-\frac{17g_2^3}{6}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{17g_2^3}{6}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\frac{31g_2^3}{6}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{7g_2^3}{6}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{31g_2^3}{6}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\frac{31g_2^3}{6}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\frac{31g_2^3}{6}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{3g_2^3}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{g_2^3}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{g_2^3}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{g_2^3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\frac{g_2^3}{6}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{5g_2^3}{6}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\frac{g_2^3}{6}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{g_2^3}{6}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\frac{g_2^3}{6}$	—

$\beta_{g_3}^{(1)}$
Quantity: g_3

$16\pi^2\,\mu\frac{dg_3}{d\mu}=\beta_{g_3}^{(1)}$

Column	Representative term structure
T1	$-7g_3^3$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-7g_3^3$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-7g_3^3$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-7g_3^3$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-7g_3^3$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-7g_3^3$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-7g_3^3$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-7g_3^3$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-7g_3^3$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-7g_3^3$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-7g_3^3$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-7g_3^3$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-7g_3^3$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-7g_3^3$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-7g_3^3$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-7g_3^3$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-7g_3^3$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-7g_3^3$

Quantity: Y_u

$$16\pi^2 \mu \frac{dY_u}{d\mu} = \beta_{Y_u}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$
T2	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$
T3	$+ 3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) \left(Y_u \right)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) \left(Y_u \right)_{ij}$

Quantity: Y_u

Terms 2 of 3

Column	Representative term structure	
T1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	
T2	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	
T3	$-8g_3^2(Y_u)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}((Y_d(Y_d)^\dagger Y_u)_{ij} - (Y_u(Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2(Y_u)_{ij}$	$-8g_3^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2 (Y_u)_{ij}$	$-8g_3^2 (Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2 (Y_u)_{ij}$	$-8g_3^2 (Y_u)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2 (Y_u)_{ij}$	$-8g_3^2 (Y_u)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{3}{2}((Y_d (Y_d)^\dagger Y_u)_{ij} - (Y_u (Y_u)^\dagger Y_u)_{ij})$	$-\frac{9}{4}g_2^2 (Y_u)_{ij}$	$-8g_3^2 (Y_u)_{ij}$

Quantity: Y_u

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{17}{12}g_1^2(Y_u)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{17}{12}g_1^2(Y_u)_{ij}$

Quantity: Y_d

$$16\pi^2 \mu \frac{dY_d}{d\mu} = \beta_{Y_d}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$\text{Tr}\left(Y_e \left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T2	$+3 \text{Tr}\left(Y_u \left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T3	$+3 \text{Tr}\left(Y_d \left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\text{Tr}\left(Y_e\left(Y_e\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_u\left(Y_u\right)^{\dagger}\right)\left(Y_d\right)_{ij}$	$+3\text{Tr}\left(Y_d\left(Y_d\right)^{\dagger}\right)\left(Y_d\right)_{ij}$

Quantity: Y_d

Terms 2 of 3

Column	Representative term structure	
T1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	
T2	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	
T3	$-8g_3^2(Y_d)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{3}{2}((Y_d(Y_d)^\dagger Y_d)_{ij} - (Y_u(Y_u)^\dagger Y_d)_{ij})$	$-\frac{9}{4}g_2^2(Y_d)_{ij}$	$-8g_3^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{2} \left((Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$	$-\frac{9}{4} g_2^2 (Y_d)_{ij}$	$-8g_3^2 (Y_d)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{2} \left((Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$	$-\frac{9}{4} g_2^2 (Y_d)_{ij}$	$-8g_3^2 (Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{2} \left((Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$	$-\frac{9}{4} g_2^2 (Y_d)_{ij}$	$-8g_3^2 (Y_d)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{2} \left((Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$	$-\frac{9}{4} g_2^2 (Y_d)_{ij}$	$-8g_3^2 (Y_d)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{2} \left((Y_d (Y_d)^\dagger Y_d)_{ij} - (Y_u (Y_u)^\dagger Y_d)_{ij} \right)$	$-\frac{9}{4} g_2^2 (Y_d)_{ij}$	$-8g_3^2 (Y_d)_{ij}$

Quantity: Y_d

Terms 3 of 3

Column	Representative term structure
T1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{5}{12}g_1^2(Y_d)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{5}{12}g_1^2(Y_d)_{ij}$

Quantity: Y_e

$$16\pi^2 \mu \frac{dY_e}{d\mu} = \beta_{Y_e}^{(1)}$$

Terms 1 of 3

Column	Representative term structure
T1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_e)_{ij}$
T2	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_e)_{ij}$
T3	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_e)_{ij}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_e)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_e)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_e)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_e)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_e)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_e)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$3 \operatorname{Tr} \left(Y_d \left(Y_d \right)^\dagger \right) (Y_e)_{ij}$	$+3 \operatorname{Tr} \left(Y_u \left(Y_u \right)^\dagger \right) (Y_e)_{ij}$	$+ \operatorname{Tr} \left(Y_e \left(Y_e \right)^\dagger \right) (Y_e)_{ij}$

Quantity: Y_e

Terms 2 of 3

Column	Representative term structure
T1	$+\frac{1}{4}(2(y_1(y_1)^\dagger Y_e)_{ij} + 3(y_2(y_2)^\dagger Y_e)_{ij} + 6(Y_e(Y_e)^\dagger Y_e)_{ij})$
T2	$-\frac{9}{4}g_2^2(Y_e)_{ij}$
T3	$-\frac{15}{4}g_1^2(Y_e)_{ij}$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_1^2(Y_e)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_1^2(Y_e)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_1^2(Y_e)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{4}g_2^2(Y_e)_{ij}$	$-\frac{15}{4}g_1^2(Y_e)_{ij}$

Quantity: Y_e

Terms 3 of 3

Column	Representative term structure
T1	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+3\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3}{4}((y_1\ (y_1)^\dagger\ Y_e)_{ij}+(y_2\ (y_2)^\dagger\ Y_e)_{ij}+2\ (Y_e\ (Y_e)^\dagger\ Y_e)_{ij})$

$$16\pi^2\,{}_{\mu}\frac{dy_1}{d\mu}=\beta_{y_1}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$2T_{\nu}^{(1)}(y_1)_{ij}$	
T2	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	
T3	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$2T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$2T_{\nu}^{(1)}(y_1)_{ij}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$2T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$2T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$2T_{\nu}^{(1)}(y_1)_{ij}$	$+\frac{3}{4}\left(y_2\,(y_2)^{\dagger}\,y_1\right)_{ij}$	$+\frac{1}{2}\left(Y_e\,(Y_e)^{\dagger}\,y_1\right)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$
T2	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$
T3	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+ \left(y_1 \left(y_1 \right)^{\dagger} y_1 \right)_{ij}$	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-\frac{9}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{33}{4} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{4} g_1^2 \left(y_1 \right)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{15}{2} g_1^2 \left(y_1 \right)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2} g_2^2 \left(y_1 \right)_{ij}$	$-\frac{3}{2} g_1^2 \left(y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{2}g_2{}^2(y_1)_{ij}$	$-\frac{3}{4}g_1{}^2(y_1)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2}g_2{}^2(y_1)_{ij}$	$-\frac{3}{2}g_1{}^2(y_1)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{2}g_2{}^2(y_1)_{ij}$	$-\frac{15}{4}g_1{}^2(y_1)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{2}g_2{}^2(y_1)_{ij}$	$-\frac{15}{2}g_1{}^2(y_1)_{ij}$

Quantity: y_1

Terms 3 of 5

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+4\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T2	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$
T3	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+4\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}(2\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	$+\frac{1}{2}((y_2\left(y_2\right)^{\dagger}y_1)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}(3\left(y_1\left(y_2\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	$+T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$T_{\nu}^{(1)}\left(y_1\right)_{ij}$	—

Terms 4 of 5

Column	Representative term structure
T1	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{i j}$
T2	$+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger} y_2\right)_{i j}+\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{i j}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{i j}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{i j}\right)$
T3	$\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{i j}$	$+\frac{1}{2}\left(2\left(y_1\left(y_2\right)^{\dagger} y_2\right)_{i j}+\left(y_2\left(y_2\right)^{\dagger} y_1\right)_{i j}+\left(Y_e\left(Y_e\right)^{\dagger} y_1\right)_{i j}+3\left(y_1\left(y_1\right)^{\dagger} y_1\right)_{i j}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$\frac{3}{2} T_{\nu}^{(1)}\left(y_1\right)_{i j}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{i j}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T2	$-\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$
T3	$+\frac{1}{2}((y_1\left(y_2\right)^{\dagger}y_2)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}\right)\left(y_2\right)_{ij}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	$1+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+5\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+\frac{1}{2}((y_1\left(y_2\right)^{\dagger}y_2)_{ij}+\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+3\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	$1+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}+\frac{1}{4}(3\left(y_2\left(y_2\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_1\right)_{ij}+6\left(y_1\left(y_1\right)^{\dagger}y_1\right)_{ij})$	—	—	—

$\beta_{y_2}^{(1)}$

Quantity: y_2

$$16\pi^2\,\mu\frac{dy_2}{d\mu}=\beta_{y_2}^{(1)}$$

Terms 1 of 5

Column		Representative term structure
T1		$T_\nu^{(2)}(y_2)_{ij}$
T2		$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$
T3		$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	$+\frac{1}{2}((y_1\ (y_1)^\dagger\ y_2)_{ij}+(Y_e\ (Y_e)^\dagger\ y_2)_{ij}+3\,(y_2\ (y_2)^\dagger\ y_2)_{ij})$	$-\frac{9}{4}g_2^2\,(y_2)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{33}{4}g_2^2\,(y_2)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$-\frac{9}{2}g_2^2\,(y_2)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$T_\nu^{(2)}(y_2)_{ij}$	—	$-\frac{9}{2}g_2^2(y_2)_{ij}$

Terms 2 of 5

Column	Representative term structure									
T1	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$									
T2	$+\frac{1}{2}((y_1\left(y_1\right)^\dagger y_2)_{ij}+(y_2\left(y_1\right)^\dagger y_1)_{ij}+(Y_e\left(Y_e\right)^\dagger y_2)_{ij}+3\left(y_2\left(y_2\right)^\dagger y_2\right)_{ij})$									
T3	$\text{Tr}(y_2\left(y_1\right)^\dagger)(y_1)_{ij}$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_1^2\left(y_2\right)_{ij}$	$+\frac{1}{2}((y_1\left(y_1\right)^\dagger y_2)_{ij}+(y_2\left(y_1\right)^\dagger y_1)_{ij}+(Y_e\left(Y_e\right)^\dagger y_2)_{ij}+3\left(y_2\left(y_2\right)^\dagger y_2\right)_{ij})$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3}{4}g_1^2\left(y_2\right)_{ij}$	—	$\text{Tr}(y_2\left(y_1\right)^\dagger)(y_1)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{15}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{15}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{15}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{15}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_1^2\left(y_2\right)_{ij}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_1^2\left(y_2\right)_{ij}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{3}{2}g_1^2\left(y_2\right)_{ij}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3}{4}g_1^2(y_2)_{ij}$	—	$\text{Tr}\left(y_2(y_1)^\dagger\right)(y_1)_{ij}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{3}{2}g_1^2(y_2)_{ij}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{15}{4}g_1^2(y_2)_{ij}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{15}{2}g_1^2(y_2)_{ij}$	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+ \frac{1}{2} ((y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (y_2 \ (y_1)^\dagger \ y_1)_{ij} + (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 3 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T2	$\frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$
T3	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{1}{2} ((y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (y_2 \ (y_1)^\dagger \ y_1)_{ij} + (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 3 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+ \frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\frac{3}{2} T_\nu^{(2)} (y_2)_{ij}$	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 5 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+ \frac{1}{4} (3 \ (y_1 \ (y_1)^\dagger \ y_2)_{ij} + 2 \ (Y_e \ (Y_e)^\dagger \ y_2)_{ij} + 6 \ (y_2 \ (y_2)^\dagger \ y_2)_{ij})$

Quantity: y_2

Terms 4 of 5

Column	Representative term structure	
T1	$\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$	
T2	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$	
T3	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)\left(y_1\right)_{ij}$	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+2\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+5\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+4\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$2T_{\nu}^{(2)}\left(y_2\right)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\frac{1}{4}\left(3\left(y_1\left(y_1\right)^{\dagger}y_2\right)_{ij}+3\left(y_2\left(y_1\right)^{\dagger}y_1\right)_{ij}+2\left(Y_e\left(Y_e\right)^{\dagger}y_2\right)_{ij}+6\left(y_2\left(y_2\right)^{\dagger}y_2\right)_{ij}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure	
T1	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	
T2	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	
T3	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3}{4} \left(y_1 \left(y_1 \right)^{\dagger} y_2 \right)_{ij}$	$+ \frac{1}{2} \left(Y_e \left(Y_e \right)^{\dagger} y_2 \right)_{ij}$	$+ \left(y_2 \left(y_2 \right)^{\dagger} y_2 \right)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$$\beta_{M_F}^{(1)}$$

Quantity: M_F

$$16\pi^2\,\mu\,\frac{dM_F}{d\mu}=\beta_{M_F}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	
T2	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	
T3	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\frac{1}{4}(2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{27}{2}g_1^2\ (M_F)_{ij}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{27}{2}g_1^2\ (M_F)_{ij}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{27}{2}g_1^2\ (M_F)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\frac{1}{4}(3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left(M_F\ (y_2)^{\dagger}\ y_2\right)_{ij})$	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{27}{2}g_1^2\ (M_F)_{ij}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{9}{2}g_2^2\ (M_F)_{ij}$	$-\frac{3}{2}g_1^2\ (M_F)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	$-\frac{3}{2}g_1^2(M_F)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{9}{2}g_2^2(M_F)_{ij}$	$-\frac{27}{2}g_1^2(M_F)_{ij}$

Quantity: M_F

Terms 2 of 5

Column	Representative term structure
T1	$\frac{1}{4}(2\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+3\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$
T2	$-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$
T3	$-6g_1^2(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\frac{1}{4}(2\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+2\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+3\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+3\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$	$-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$-6g_1^2(M_F)_{ij}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$-6g_1^2(M_F)_{ij}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-6g_1^2(M_F)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\frac{1}{4}(3\left((M_F)^T\ (y_1)^\dagger\ y_1\right)_{ij}+3\left((y_1)^T\ (y_1)^*\ M_F\right)_{ij}+2\left((M_F)^T\ (y_2)^\dagger\ y_2\right)_{ij}+2\left((y_2)^T\ (y_2)^*\ M_F\right)_{ij})$	$-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-6g_1^2(M_F)_{ij}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{9}{4}g_2^2((M_F)^T)_{ij}+(M_F)_{ij})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$-6g_1^2(M_F)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: M_F

Terms 3 of 5

Column	Representative term structure
T1	$\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$
T2	$+ \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$
T3	$\left((M_F)^T \ (y_1)^\dagger \ y_1 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$	$+ \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$	—	$\left((M_F)^T \ (y_1)^\dagger \ y_1 \right)_{ij}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$	$+ \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$	$+ \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\left((y_1)^T \ (y_1)^* \ M_F \right)_{ij}$	$+ \left(M_F \ (y_2)^\dagger \ y_2 \right)_{ij}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+ \left((M_F)^T (y_2)^\dagger y_2 \right)_{ij}$
T2	$+ \left((y_2)^T (y_2)^* M_F \right)_{ij}$
T3	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \left((M_F)^T (y_2)^\dagger y_2 \right)_{ij}$	$+ \left((y_2)^T (y_2)^* M_F \right)_{ij}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$\frac{1}{2} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$\frac{3}{4} \left((y_1)^T (y_1)^* M_F \right)_{ij} + \left(M_F (y_2)^\dagger y_2 \right)_{ij}$

Quantity: M_F

Terms 5 of 5

Column	Representative term structure
T1	$-12g_2^2\left(M_F\right)_{ij}$
T2	$\frac{1}{2}((M_F)^T\left(y_1\right)^{\dagger}y_1)_{ij}+\left((y_1)^T\left(y_1\right)^{*}M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^{*}M_F\right)_{ij}$
T3	$-6g_2^2((M_F)^T)_{ij}+(M_F)_{ij}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-12g_2^2\left(M_F\right)_{ij}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$\frac{1}{2}((M_F)^T\left(y_1\right)^{\dagger}y_1)_{ij}+\left((y_1)^T\left(y_1\right)^{*}M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^{*}M_F\right)_{ij}$	$-6g_2^2((M_F)^T)_{ij}+(M_F)_{ij}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12g_2^2\left(M_F\right)_{ij}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-12g_2^2\left(M_F\right)_{ij}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-12g_2^2\left(M_F\right)_{ij}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\frac{3}{4}((M_F)^T\left(y_1\right)^{\dagger}y_1)_{ij}+\left((y_1)^T\left(y_1\right)^{*}M_F\right)_{ij}+\left((M_F)^T\left(y_2\right)^{\dagger}y_2\right)_{ij}+\left((y_2)^T\left(y_2\right)^{*}M_F\right)_{ij}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$\beta_{m_1^2}^{(1)}$

Quantity: m_1^2

$16\pi^2\,\mu\frac{dm_1^2}{d\mu}=\beta_{m_1^2}^{(1)}$

Terms 1 of 5

Column	Representative term structure
T1	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$
T2	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$
T3	$-6g_1^2m_1^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$m_1^2(4T_\nu^{(1)}+4\lambda_{S_1}^{(1)})$	$-8\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	$-6g_1^2m_1^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(M_F)^\dagger\,(y_1)^T\,(y_1)^*\Big)$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-4 \operatorname{Tr} \left(M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-4 \operatorname{Tr} \left(M_F (M_F)^\dagger (y_1)^T (y_1)^* \right)$	$-6 g_1^2 m_1^2$

Quantity: m_1^2

Terms 2 of 5

Column	Representative term structure
T1	$+6\lambda_{12}^{(1)}m_2^2$
T2	$-16\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$
T3	$-16\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(1)}m_2^2$	$-16\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-16\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(1)}m_2^2$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)}m_2^2$	$-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)}m_2^2$	$-12\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-12\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}m_2^2$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}m_2^2$	$-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}m_2^2$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-8\text{Tr}\left(M_F\left(M_F\right)^*\left(y_1\right)^T\left(y_1\right)^*\right)$	$-8\text{Tr}\left(M_F\left(y_1\right)^\dagger y_1\left(M_F\right)^*\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: m_1^2

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{3g_1^2m_1^2}{2}$
T2	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$
T3	$-\frac{9g_2^2m_1^2}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_1^2m_1^2}{2}$	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{3g_1^2m_1^2}{2}$	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$m_1^2(2T_\nu^{(1)}+6\lambda_{S_1}^{(1)})$	$-\frac{9g_2^2m_1^2}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3g_1^2m_1^2}{2}$	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{3g_1^2m_1^2}{2}$	—	$-\frac{9g_2^2m_1^2}{2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{9g_2^2m_1^2}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_1^2}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{3g_1^2 m_1^2}{2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: m_1^2

Terms 4 of 5

Column	Representative term structure	
T1	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	
T2	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	
T3	$-12g_2^2m_1^2$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$m_1^2(3T_\nu^{(1)} + 6\lambda_{S_1}^{(1)})$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$m_1^2(2T_\nu^{(1)} + 8\lambda_{S_1}^{(1)} + 4\lambda_{S_1}^{(A)})$	$-12g_2^2m_1^2$

Quantity: m_1^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+m_2^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$$\beta_{m_2^2}^{(1)}$$

Quantity: m_2^2

$$16\pi^2\,\mu\frac{dm_2^2}{d\mu}=\beta_{m_2^2}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$
T2	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$
T3	$-12g_2^{\,2}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	$-12g_2^{\,2}m_2^2$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-6\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-8\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	—	$-12g_2^{\,2}m_2^2$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$m_2^2(2T_\nu^{(2)}+8\lambda_{S_2}^{(1)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\Big(M_F\,(y_2)^\dagger\,y_2\,(M_F)^\dagger\Big)$	$-12g_2^{\,2}m_2^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$	$-12g_2^2 m_2^2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$m_2^2(2T_\nu^{(2)} + 8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})$	$-4 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^\dagger \right)$	$-12g_2^2 m_2^2$

Quantity: m_2^2

Terms 2 of 5

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)} m_1^2$
T2	$-8 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$
T3	$-8 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)} m_1^2$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)} m_1^2$	$-8 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$	$-8 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)} m_1^2$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)} m_1^2$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)} m_1^2$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)} m_1^2$	$-8 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$	$-8 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)} m_1^2$	$-12 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$	$-12 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}^{(1)} m_1^2$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)} m_1^2$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(1)} m_1^2$	$-16 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$	$-16 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(1)} m_1^2$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(1)} m_1^2$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(1)} m_1^2$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-8 \text{Tr} \left(M_F (M_F)^* (y_2)^T (y_2)^* \right)$	$-8 \text{Tr} \left(M_F (y_2)^\dagger y_2 (M_F)^* \right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: m_2^2

Terms 3 of 5

Column	Representative term structure
T1	$-\frac{3g_1^2m_2^2}{2}$
T2	$-6g_1^2m_2^2$
T3	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_2^2}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$-6g_1^2m_2^2$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{27g_1^2m_2^2}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$-24g_1^2m_2^2$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{3g_1^2m_2^2}{2}$	—	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-6g_1^2m_2^2$	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{27g_1^2m_2^2}{2}$	—	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-24g_1^2m_2^2$	$m_2^2(2T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{3g_1^2m_2^2}{2}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-6g_1^2m_2^2$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{27g_1^2m_2^2}{2}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-24g_1^2m_2^2$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_2^2}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-6g_1^2m_2^2$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{27g_1^2m_2^2}{2}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-24g_1^2m_2^2$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{3g_1^2m_2^2}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-6g_1^2m_2^2$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{27g_1^2m_2^2}{2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-24g_1^2m_2^2$	—

Quantity: m_2^2

Terms 4 of 5

Column		Representative term structure
T1		$-\frac{9g_2^2m_2^2}{2}$
T2		$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$
T3		$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-\frac{9g_2^2m_2^2}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{9g_2^2m_2^2}{2}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-\frac{9g_2^2m_2^2}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-\frac{9g_2^2m_2^2}{2}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-\frac{9g_2^2m_2^2}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-\frac{9g_2^2m_2^2}{2}$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{9g_2^2m_2^2}{2}$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{9g_2^2m_2^2}{2}$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{9g_2^2m_2^2}{2}$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{9g_2^2m_2^2}{2}$	$m_2^2(3T_\nu^{(2)}+6\lambda_{S_2}^{(1)})$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$m_2^2(4T_\nu^{(2)}+4\lambda_{S_2}^{(1)})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: m_2^2

Terms 5 of 5

Column	Representative term structure
T1	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+m_1^2(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Quantity: λ_1

$$16\pi^2 \mu \frac{d\lambda_1}{d\mu} = \beta_{\lambda_1}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_1 (12 \text{Tr} (Y_d (Y_d)^\dagger) + 4 \text{Tr} (Y_e (Y_e)^\dagger) + 12 \text{Tr} (Y_u (Y_u)^\dagger))$
T2	$-12 \text{Tr} (Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$
T3	$-4 \text{Tr} (Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_1(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$	$-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$	$-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_1(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$	$-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$	$-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_1(12 \text{Tr}(Y_d (Y_d)^\dagger) + 4 \text{Tr}(Y_e (Y_e)^\dagger) + 12 \text{Tr}(Y_u (Y_u)^\dagger))$	$-12 \text{Tr}(Y_d (Y_d)^\dagger Y_d (Y_d)^\dagger)$	$-4 \text{Tr}(Y_e (Y_e)^\dagger Y_e (Y_e)^\dagger)$

Terms 2 of 7

Column	Representative term structure
T1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$
T2	$+4 \lambda_5 \left(\lambda_5\right)^{*}$
T3	$+\frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-12 \operatorname{Tr}\left(Y_u \left(Y_u\right)^{\dagger} Y_u \left(Y_u\right)^{\dagger}\right)$	$+4 \lambda_5 \left(\lambda_5\right)^{*}$	$+\frac{9 g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$	$+\frac{9g_2^4}{4}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$	$+\frac{9g_2^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$	$+\frac{9g_2^4}{4}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$	$+\frac{9g_2^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-12 \text{Tr} \left(Y_u (Y_u)^\dagger Y_u (Y_u)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$	$+\frac{9g_2^4}{4}$

Quantity: λ_1

Terms 3 of 7

Column	Representative term structure
T1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$
T2	$-9g_2^2\lambda_1$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_1\right)$	$-9g_2^2\lambda_1$	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2 \left(\frac{3g_2^2}{2} - 3\lambda_1 \right)$	$-9g_2^2 \lambda_1$	$+\frac{3g_1^4}{4}$

Terms 4 of 7

Column	Representative term structure
T1	$+12\lambda_1^2$
T2	$+2\lambda_{HS_1}^{(1)2}$
T3	$+6\lambda_{HS_2}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+12\lambda_1^2$	$+2\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+12\lambda_1^2$	$+4\lambda_{HS_1}^{(1)2}$	$+4\lambda_{HS_2}^{(1)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+2\lambda_{HS_2}^{(1)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+12\lambda_1^2$	$+6\lambda_{HS_1}^{(1)2}$	$+6\lambda_{HS_2}^{(1)2}$

Quantity: λ_1

Terms 5 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_2}^{(A)^2}$
T2	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$
T3	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+\frac{\lambda_{HS_1}^{(A)^2}}{4}$	$+\frac{\lambda_{HS_2}^{(A)^2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{HS_2}^{(A)^2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{HS_2}^{(A)^2}$	—	—

Terms 6 of 7

Column	Representative term structure
T1	$+8\lambda_7 (\lambda_7)^*$
T2	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3	$+8\lambda_8 (\lambda_8)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_7 (\lambda_7)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+8\lambda_8 (\lambda_8)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_7 (\lambda_7)^*$	$+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$+8\lambda_8 (\lambda_8)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 7 of 7

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+3\lambda_{HS_1}^{(A)^2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+3\lambda_{HS_1}^{(A)^2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{HS_1}^{(A)^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{HS_1}^{(A)^2}$

$\beta_{\lambda_{S_1}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{S_1}^{(1)}}{d\mu}=\beta_{\lambda_{S_1}^{(1)}}^{(1)}$$

Terms 1 of 10

Column	Representative term structure
T1	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$
T2	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$
T3	$+12g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$8\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-8\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$4\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-4\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$6\lambda_{S_1}^{(1)}T_\nu^{(1)}$	$-5\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-3\text{Tr}\left(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\right)$	$+12g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-3 \operatorname{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-3 \operatorname{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-3 \operatorname{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-3 \operatorname{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$	$+12g_1^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 2 of 10

Column	Representative term structure
T1	$-12g_1^2\lambda_{S_1}^{(1)}$
T2	$+6\lambda_{12}^{(1)2}$
T3	$+4\lambda_{HS_1}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-3g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12g_1^2\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+4\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+2\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_{12}^{(1)2}$	$+4\lambda_{HS_1}^{(1)2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 3 of 10

Column	Representative term structure
T1	$+10\lambda_{S_1}^{(1)2}$
T2	$+\frac{3g_1^4}{4}$
T3	$+\frac{9g_2^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+10\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+10\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+10\lambda_{S_1}^{(1)2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+12\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	$+\frac{9g_2^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+12\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	$+\frac{9g_2^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+12\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	$+\frac{9g_2^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+12\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	$+\frac{9g_2^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+12\lambda_{S_1}^{(1)2}$	—	$+\frac{9g_2^4}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+14\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+14\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+14\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+14\lambda_{S_1}^{(1)2}$	$+\frac{3g_1^4}{4}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+14\lambda_{S_1}^{(1)2}$	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 4 of 10

Column	Representative term structure
T1	$+g_1^2\left(6g_2^2-12\lambda_{S_1}^{(1)}\right)$
T2	$-9g_2^2\lambda_{S_1}^{(1)}$
T3	$+\frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_1}^{(1)}\right)$	$-9g_2^2\lambda_{S_1}^{(1)}$	$+\frac{\lambda_{12}^{(A)2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 5 of 10

Column	Representative term structure
T1	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$
T2	$+\lambda_{11}(2\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)+24\,(\lambda_{11})^*)$
T3	$+2\,(\lambda_{11})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+\lambda_{11}(2\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)+24\,(\lambda_{11})^*)$	$+2\,(\lambda_{11})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	$+\lambda_{11}(3\text{Tr}\Big(y_1\ (y_2)^\dagger\Big)+24\,(\lambda_{11})^*)$	$+3\,(\lambda_{11})^*\text{Tr}\Big(y_2\ (y_1)^\dagger\Big)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{\lambda_{HS_1}^{(A) \ 2}}{4}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 6 of 10

Column	Representative term structure									
T1	$+8\lambda_7\left(\lambda_7\right)^*$									
T2	$+8\lambda_9\left(\lambda_9\right)^*$									
T3	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_7\left(\lambda_7\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_1}^{(1)}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_1}^{(1)}$

Terms 7 of 10

Column	Representative term structure
T1	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$
T2	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$
T3	$+21g_2^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_1}^{(1)}(4T_\nu^{(1)} + 14\lambda_{S_1}^{(A)})$	$+2\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$+21g_2^4$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 8 of 10

Column	Representative term structure	
T1	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	
T2	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	
T3	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_1}^{(1)}-6\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+g_2^2(-24\lambda_{S_1}^{(1)}-12\lambda_{S_1}^{(A)})$	$+\frac{3\lambda_{HS_1}^{(A)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2 \left(12g_2^2 - 12\lambda_{S_1}^{(1)} - 6\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-24\lambda_{S_1}^{(1)} - 12\lambda_{S_1}^{(A)})$	$+ \frac{3\lambda_{HS_1}^{(A)2}}{2}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 9 of 10

Column	Representative term structure
T1	$+26\lambda_{S_1}^{(A)2}$
T2	$+g_1^2\left(3g_2^2-3\lambda_{S_1}^{(1)}-\frac{3\lambda_{S_1}^{(A)}}{2}\right)$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+26\lambda_{S_1}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_1}^{(1)}-\frac{3\lambda_{S_1}^{(A)}}{2}\right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+26\lambda_{S_1}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_1}^{(1)}-\frac{3\lambda_{S_1}^{(A)}}{2}\right)$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_1}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_1}^{(1)}-\frac{3\lambda_{S_1}^{(A)}}{2}\right)$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_1}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_1}^{(1)}-\frac{3\lambda_{S_1}^{(A)}}{2}\right)$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_1}^{(A)2}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_1}^{(1)}$

Terms 10 of 10

Column	Representative term structure
T1	$+6\lambda_{12}^{(A)2}$
T2	$+6\lambda_{12}^{(\times)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(1)}$

$16\pi^2\mu\frac{d\lambda_{S_2}^{(1)}}{d\mu}=\beta_{\lambda_{S_2}^{(1)}}^{(1)}$

Terms 1 of 10

Column	Representative term structure
T1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$
T2	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$
T3	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-5\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$-8\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)}+14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2(y_2)^\dagger y_2(y_2)^\dagger\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2(y_2)^\dagger y_2(y_2)^\dagger\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2(y_2)^\dagger y_2(y_2)^\dagger\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}^{(1)}(4T_\nu^{(2)} + 14\lambda_{S_2}^{(A)})$	$+2\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-3\text{Tr}\left(y_2(y_2)^\dagger y_2(y_2)^\dagger\right)$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 2 of 10

Column	Representative term structure
T1	$+21g_2^4$
T2	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$
T3	$+2\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+2\lambda_{12}^{(1)2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+4\lambda_{12}^{(1)2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+4\lambda_{12}^{(1)2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+4\lambda_{12}^{(1)2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+4\lambda_{12}^{(1)2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+6\lambda_{12}^{(1)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$+6\lambda_{12}^{(1)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$+6\lambda_{12}^{(1)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$+6\lambda_{12}^{(1)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+6\lambda_{12}^{(1)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+6\lambda_{12}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+6\lambda_{12}^{(1)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+6\lambda_{12}^{(1)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+21g_2^4$	$+g_2^2(-24\lambda_{S_2}^{(1)}-12\lambda_{S_2}^{(A)})$	$+6\lambda_{12}^{(1)2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 3 of 10

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1) \ 2}$
T2	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$
T3	$+14\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+12\lambda_{S_2}^{(1) \ 2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	—	$+10\lambda_{S_2}^{(1) \ 2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1) \ 2}$	$+\frac{3\lambda_{HS_2}^{(A) \ 2}}{2}$	$+14\lambda_{S_2}^{(1) \ 2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1)2}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{2}$	$+14\lambda_{S_2}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)2}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{2}$	$+14\lambda_{S_2}^{(1)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1)2}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{2}$	$+14\lambda_{S_2}^{(1)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)2}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{2}$	$+14\lambda_{S_2}^{(1)2}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 4 of 10

Column	Representative term structure
T1	$+26\lambda_{S_2}^{(A)2}$
T2	$+g_1^2\left(3g_2^2-3\lambda_{S_2}^{(1)}-\frac{3\lambda_{S_2}^{(A)}}{2}\right)$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_2}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_2}^{(1)}-\frac{3\lambda_{S_2}^{(A)}}{2}\right)$	$+\frac{3g_1^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_2}^{(A)2}$	$+g_1^2\left(27g_2^2-27\lambda_{S_2}^{(1)}-\frac{27\lambda_{S_2}^{(A)}}{2}\right)$	$+\frac{243g_1^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+\frac{3g_1^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+\frac{243g_1^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{3g_1^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+\frac{243g_1^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$+\frac{3g_1^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$+\frac{243g_1^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+26\lambda_{S_2}^{(A)2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+26\lambda_{S_2}^{(A)2}$	$+g_1^2\left(3g_2^2-3\lambda_{S_2}^{(1)}-\frac{3\lambda_{S_2}^{(A)}}{2}\right)$	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+26\lambda_{S_2}^{(A)^2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+26\lambda_{S_2}^{(A)^2}$	$+g_1^2 \left(27g_2^2 - 27\lambda_{S_2}^{(1)} - \frac{27\lambda_{S_2}^{(A)}}{2} \right)$	$+\frac{243g_1^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+26\lambda_{S_2}^{(A)^2}$	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 5 of 10

Column	Representative term structure
T1	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$
T2	$+12g_1^4$
T3	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+12g_1^4$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_1^2\left(48g_2^2-48\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)}\right)$	$+192g_1^4$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+12g_1^4$	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+192g_1^4$	$4\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+12g_1^4$	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+192g_1^4$	$6\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+12g_1^4$	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+192g_1^4$	$8\lambda_{S_2}^{(1)}T_\nu^{(2)}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-12\lambda_{S_2}^{(1)}-6\lambda_{S_2}^{(A)}\right)$	$+12g_1^4$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2\left(48g_2^2-48\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)}\right)$	$+192g_1^4$	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 6 of 10

Column	Representative term structure
T1	$+ \frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{S_2}^{(1)}$
T3	$+ \frac{\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+ \frac{9g_2^4}{4}$	$-9g_2^2\lambda_{S_2}^{(1)}$	$+ \frac{\lambda_{12}^{(A)2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 7 of 10

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$
T2	$\lambda_{10}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$
T3	$+2\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$\lambda_{10}(2\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$	$+2\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	$\lambda_{10}(3\text{Tr}\left(y_1\ (y_2)^\dagger\right)+24\ (\lambda_{10})^*)$	$+3\ (\lambda_{10})^*\text{Tr}\left(y_2\ (y_1)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{\lambda_{HS_2}^{(A) \ 2}}{4}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 8 of 10

Column	Representative term structure	
T1	$+8\lambda_8\left(\lambda_8\right)^*$	
T2	$+8\lambda_9\left(\lambda_9\right)^*$	
T3	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_8\left(\lambda_8\right)^*$	$+8\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(\frac{3g_2^2}{2}-3\lambda_{S_2}^{(1)}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_1^2\left(\frac{27g_2^2}{2}-27\lambda_{S_2}^{(1)}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{S_2}^{(1)}$

Terms 9 of 10

Column	Representative term structure
T1	$+g_1^2\left(6g_2^2-12\lambda_{S_2}^{(1)}\right)$
T2	$-3g_1^2\lambda_{S_2}^{(1)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_2}^{(1)}\right)$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_1^2\left(24g_2^2-48\lambda_{S_2}^{(1)}\right)$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_1^2\left(6g_2^2-12\lambda_{S_2}^{(1)}\right)$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_1^2\left(24g_2^2-48\lambda_{S_2}^{(1)}\right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-3g_1^2\lambda_{S_2}^{(1)}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-12g_1^2\lambda_{S_2}^{(1)}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-27g_1^2\lambda_{S_2}^{(1)}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-48g_1^2\lambda_{S_2}^{(1)}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_2}^{(1)}$

Terms 10 of 10

Column	Representative term structure
T1	$+6\lambda_{12}^{(A)2}$
T2	$+6\lambda_{12}^{(\times)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(A)2}$	$+6\lambda_{12}^{(\times)2}$

$$\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Quantity: $\lambda_{HS_1}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(1)}}{d\mu}=\beta_{\lambda_{HS_1}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$
T3	$+12\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_1}^{(1)}(4T_\nu^{(1)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+12\lambda_5\,(\lambda_5)^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-3\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-3\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-3\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-3\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-3\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+6\lambda_5\,(\lambda_5)^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,Y_e\,(Y_e)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-2 \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$+4\lambda_5 (\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$
T2	$+3g_1^4$
T3	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{9g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	$-\frac{3g_1^2\lambda_{HS_1}^{(1)}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(1)}}{2}$	$+3g_1^4$	$-\frac{15g_1^2\lambda_{HS_1}^{(1)}}{2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 3 of 9

Column		Representative term structure
T1		$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$
T2		$+6\lambda_1\lambda_{HS_1}^{(1)}$
T3		$+4\lambda_{HS_1}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)}$	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_1\lambda_{HS_1}^{(1)}$	$+4\lambda_{HS_1}^{(1)^2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+\frac{3g_1^4}{4}$
T3	$-3g_1^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	$+\frac{3g_1^4}{4}$	$-3g_1^2\lambda_{HS_1}^{(1)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}$	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\frac{9g_2^4}{4}$
T3	$-9g_2^2\lambda_{HS_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{HS_1}^{(1)}(2T_\nu^{(1)}+6\operatorname{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\operatorname{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\operatorname{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_1}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)$
T3	$+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{11}\right)^*\right)+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{3\lambda_{HS_1}^{(A)^2}}{4}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$+24\lambda_7 (\lambda_7)^*$
T2	$+3\lambda_{11} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T3	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_7 (\lambda_7)^*$	$+3\lambda_{11} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_7 (\lambda_7)^*$	$+3\lambda_{11} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$\lambda_{HS_1}^{(1)} (3T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 8 of 9

Column	Representative term structure
T1	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$
T2	$+6g_2^4$
T3	$+6\lambda_{HS_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_1}^{(1)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_1}^{(A)})$	$+6g_2^4$	$+6\lambda_{HS_1}^{(A)2}$

Quantity: $\lambda_{HS_1}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\lambda_{HS_2}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$$\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(1)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(1)}}{d\mu}=\beta_{\lambda_{HS_2}^{(1)}}^{(1)}$$

Terms 1 of 9

Column	Representative term structure
T1	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$
T2	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+4\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-3\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-3\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-3\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-3\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-3\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+6\lambda_5\ (\lambda_5)^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+12\lambda_5\ (\lambda_5)^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-4\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+12\lambda_5\ (\lambda_5)^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+12\lambda_5\ (\lambda_5)^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+12\lambda_5\ (\lambda_5)^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+12\lambda_5\ (\lambda_5)^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+4\lambda_{S_2}^{(A)})$	$-2\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+4\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	$-2 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	$-2 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	$-2 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$	$+4\lambda_5 (\lambda_5)^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_2}^{(1)}(2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 4\lambda_{S_2}^{(A)})$	$-2 \text{Tr}(y_2 (y_2)^\dagger Y_e (Y_e)^\dagger)$	$+4\lambda_5 (\lambda_5)^*$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 2 of 9

Column	Representative term structure
T1	$+6g_2^4$
T2	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$
T3	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{9g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{15g_1^2\lambda_{HS_2}^{(1)}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+6g_2^4$	$-\frac{33g_2^2\lambda_{HS_2}^{(1)}}{2}$	$-\frac{51g_1^2\lambda_{HS_2}^{(1)}}{2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 3 of 9

Column		Representative term structure
T1		$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$
T2		$+6\lambda_1\lambda_{HS_2}^{(1)}$
T3		$+4\lambda_{HS_2}^{(1)^2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+6\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)}$	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+6\lambda_1\lambda_{HS_2}^{(1)}$	$+4\lambda_{HS_2}^{(1)^2}$

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 4 of 9

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$
T2	$+6\lambda_{HS_2}^{(A)\ 2}$
T3	$+\frac{3g_1^4}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	$+\frac{3g_1^4}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	$+\frac{27g_1^4}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{3g_1^4}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{27g_1^4}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{3g_1^4}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{27g_1^4}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{3g_1^4}{4}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	$+\frac{27g_1^4}{4}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	$+\frac{3g_1^4}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)\ 2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)~2}$	$+\frac{27g_1^4}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)}$	$+6\lambda_{HS_2}^{(A)~2}$	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 5 of 9

Column	Representative term structure
T1	$-3g_1^2 \lambda_{HS_2}^{(1)}$
T2	$+3g_1^4$
T3	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	$+3g_1^4$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-15g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$+12g_1^4$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_1^2 \lambda_{HS_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$+3g_1^4$	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-15g_1^2 \lambda_{HS_2}^{(1)}$	—	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$+12g_1^4$	$\lambda_{HS_2}^{(1)} (2T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+3g_1^4$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-15g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+12g_1^4$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-3g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+3g_1^4$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-15g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$+12g_1^4$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-3g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+3g_1^4$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-15g_1^2 \lambda_{HS_2}^{(1)}$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+12g_1^4$	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 6 of 9

Column	Representative term structure
T1	$+\frac{9g_2^4}{4}$
T2	$-9g_2^2\lambda_{HS_2}^{(1)}$
T3	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{9g_2^4}{4}$	$-9g_2^2\lambda_{HS_2}^{(1)}$	$+\frac{3\lambda_{HS_2}^{(A)2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 7 of 9

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(\frac{1}{2} \text{Tr}(y_1 (y_2)^\dagger) + 4 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + 3(\lambda_{10})^* \right)$
T2	$+\frac{1}{2} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$
T3	$+3\lambda_{10} \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{1}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)+\frac{1}{2}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$		$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+4\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+3\left(\lambda_{10}\right)^*\right)+\frac{3}{4}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$		$+3\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 8 of 9

Column	Representative term structure
T1	$+24\lambda_8\left(\lambda_8\right)^*$
T2	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_8\left(\lambda_8\right)^*$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_8\left(\lambda_8\right)^*$	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\lambda_{HS_2}^{(1)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$\lambda_{HS_2}^{(1)}\left(4T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(1)}$

Terms 9 of 9

Column	Representative term structure
T1	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\lambda_{HS_1}^{(1)}(6\lambda_{12}^{(1)}+4\lambda_{12}^{(\times)})$

$\beta_{\lambda_{12}^{(1)}}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

$16\pi^2\,\mu\frac{d\lambda_{12}^{(1)}}{d\mu}=\beta_{\lambda_{12}^{(1)}}^{(1)}$

Terms 1 of 12

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$
T2	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$
T3	$+4\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{12}^{(1)}(4T_{\nu}^{(1)}+2T_{\nu}^{(2)}+4\lambda_{S_2}^{(A)})$	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+3\lambda_5\,(\lambda_5)^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+3\lambda_5\,(\lambda_5)^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-4\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+4\lambda_5\,(\lambda_5)^*$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-2\,\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-2 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-2 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_2 \right)^\dagger \right)$	—

Quantity: $\lambda_{12}^{(1)}$

Terms 2 of 12

Column	Representative term structure
T1	$-12g_2^2\lambda_{12}^{(1)}$
T2	$-6g_1^2\lambda_{12}^{(1)}$
T3	$+4\lambda_{12}^{(1)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-12g_2^2\lambda_{12}^{(1)}$	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-12g_2^2\lambda_{12}^{(1)}$	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}^{(1)}$	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_{12}^{(1)}$	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}^{(1)}$	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{12}^{(1)}$	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{(1)}$	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-12g_2^2\lambda_{12}^{(1)}$	$-3g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-6g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-12g_2^2\lambda_{12}^{(1)}$	$-15g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-12g_2^2\lambda_{12}^{(1)}$	$-30g_1^2\lambda_{12}^{(1)}$	$+4\lambda_{12}^{(1)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+4\lambda_{12}^{(1)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{12}^{(1)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+4\lambda_{12}^{(1)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+4\lambda_{12}^{(1)2}$

Quantity: $\lambda_{12}^{(1)}$

Terms 3 of 12

Column	Representative term structure
T1	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$
T2	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+6\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+6\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+8\lambda_{12}^{(1)}\lambda_{S_1}^{(1)}$	$+4\lambda_{12}^{(1)}\lambda_{S_2}^{(1)}$	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{12}^{(1)}$

Terms 4 of 12

Column	Representative term structure
T1	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+\frac{3g_1^4}{4}$
T3	$+48g_1^4$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{4}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_1^4}{4}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	$+48g_1^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{4}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$+\frac{27g_1^4}{4}$	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+48g_1^4$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{4}$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{27g_1^4}{4}$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+48g_1^4$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-4\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{4}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$+\frac{27g_1^4}{4}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$+48g_1^4$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-2\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{4}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_1^4}{4}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+48g_1^4$

Quantity: $\lambda_{12}^{(1)}$

Terms 5 of 12

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$
T2	$+\frac{9g_2^4}{4}$
T3	$+\frac{3\lambda_{12}^{(A)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{12}^{(1)}(2T_\nu^{(1)} + 2T_\nu^{(2)})$	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$+\frac{9g_2^4}{4}$	$+\frac{3\lambda_{12}^{(A)2}}{4}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 6 of 12

Column	Representative term structure
T1	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left((\lambda_{10})^*+(\lambda_{11})^*\right)\right)$
T2	$+\frac{3}{2}\left((\lambda_{10})^*+(\lambda_{11})^*\right)\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left((\lambda_{10})^*+(\lambda_{11})^*\right)\right)$ $+\lambda_{11}\left(\frac{3}{2}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left((\lambda_{10})^*+(\lambda_{11})^*\right)\right)$	$+\frac{3}{2}\left((\lambda_{10})^*+(\lambda_{11})^*\right)\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{10}\left(\frac{9}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left((\lambda_{10})^*+(\lambda_{11})^*\right)\right)$ $+\lambda_{11}\left(\frac{9}{4}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)+9\left((\lambda_{10})^*+(\lambda_{11})^*\right)\right)$	$+\frac{9}{4}\left((\lambda_{10})^*+(\lambda_{11})^*\right)\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 7 of 12

Column	Representative term structure
T1	$+24\lambda_9\left(\lambda_9\right)^*$
T2	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$
T3	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+24\lambda_9\left(\lambda_9\right)^*$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+24\lambda_9\left(\lambda_9\right)^*$	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\lambda_{12}^{(1)}\left(3T_{\nu}^{(1)}+3T_{\nu}^{(2)}\right)$	$-\frac{9}{2}\text{Tr}\left(y_1\left(y_1\right)^{\dagger}y_2\left(y_2\right)^{\dagger}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{12}^{(1)}$

Terms 8 of 12

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$
T2	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)$
T3	$+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{12}^{(1)}(2T_{\nu}^{(1)}+4T_{\nu}^{(2)}+4\lambda_{S_1}^{(A)})$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$\lambda_{12}^{(1)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}+4\lambda_{S_1}^{(A)}+4\lambda_{S_2}^{(A)}\right)+\lambda_{12}^{(\times)}\left(\frac{4}{3}T_{\nu}^{(1)}+\frac{4}{3}T_{\nu}^{(2)}\right)$	

Quantity: $\lambda_{12}^{(1)}$

Terms 9 of 12

Column	Representative term structure
T1	$+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$
T2	$+16g_2^4$
T3	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$		$+16g_2^4$	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	$0+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$		$+16g_2^4$	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$		$+16g_2^4$	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	$1+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$		$+16g_2^4$	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}+\frac{4}{3}\lambda_5\left(\lambda_5\right)^*$		$+16g_2^4$	$+g_2^2\left(-24\lambda_{12}^{(1)}-16\lambda_{12}^{(\times)}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 10 of 12

Column	Representative term structure	
T1	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	
T2	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	
T3	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_1^2(-3\lambda_{12}^{(1)}-2\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2(-6\lambda_{12}^{(1)}-4\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2(-15\lambda_{12}^{(1)}-10\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2(-30\lambda_{12}^{(1)}-20\lambda_{12}^{(\times)})$	$+\lambda_{S_1}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$	$+\lambda_{S_2}^{(1)}\left(8\lambda_{12}^{(1)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Quantity: $\lambda_{12}^{(1)}$

Terms 11 of 12

Column	Representative term structure
T1	$+16\lambda_{12}^{(A)2}$
T2	$+\frac{32\lambda_{12}^{(\times)2}}{3}$
T3	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$

Model	α	d_{S1}	Y_{S1}	d_{S2}	Y_{S2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+16\lambda_{12}^{(A)2}$	$+\frac{32\lambda_{12}^{(\times)2}}{3}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S1}^{(A)}}{3}$

Quantity: $\lambda_{12}^{(1)}$

Terms 12 of 12

Column	Representative term structure
T1	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{8\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}}{3}$

$$\beta_{\lambda_5}^{(1)}$$

Quantity: λ_5

$$16\pi^2\,\mu\,\frac{d\lambda_5}{d\mu}=\beta_{\lambda_5}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$
T2	$-\frac{21g_2^2\lambda_5}{2}$
T3	$-\frac{9g_1^2\lambda_5}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_5(2T_\nu^{(1)}+T_\nu^{(2)}+6\,\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\,\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\,\text{Tr}\big(Y_u\,(Y_u)^\dagger\big)+4\lambda_{HS_2}^{(A)})$	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	$-\frac{9g_1^2\lambda_5}{2}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{9g_1^2\lambda_5}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$-\frac{33g_1^2\lambda_5}{2}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-\frac{9g_1^2\lambda_5}{2}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{9g_1^2\lambda_5}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$-\frac{33g_1^2\lambda_5}{2}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{21g_2^2\lambda_5}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{21g_2^2\lambda_5}{2}$	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{21g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{33g_2^2\lambda_5}{2}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{9g_1^2\lambda_5}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{33g_2^2\lambda_5}{2}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{33g_2^2\lambda_5}{2}$	$-\frac{33g_1^2\lambda_5}{2}$

Quantity: λ_5

Terms 2 of 6

Column	Representative term structure
T1	$+2\lambda_{12}^{(1)}\lambda_5$
T2	$+2\lambda_1\lambda_5$
T3	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(1)}\lambda_5$	$+2\lambda_1\lambda_5$	$+4\lambda_{HS_1}^{(1)}\lambda_5$

Quantity: λ_5

Terms 3 of 6

Column	Representative term structure
T1	$+4\lambda_{HS_2}^{(1)}\lambda_5$
T2	$-3g_1^2\lambda_5$
T3	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-3g_1^2\lambda_5$	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-9g_1^2\lambda_5$	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-3g_1^2\lambda_5$	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-9g_1^2\lambda_5$	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	$\lambda_5\left(T_\nu^{(1)}+T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+\frac{\lambda_{12}^{(A)}}{6}+\frac{2\lambda_{HS_1}^{(A)}}{3}+2\lambda_{HS_2}^{(A)}\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-3g_1^2\lambda_5$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-9g_1^2\lambda_5$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-3g_1^2\lambda_5$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-9g_1^2\lambda_5$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-3g_1^2\lambda_5$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+4\lambda_{HS_2}^{(1)}\lambda_5$	$-9g_1^2\lambda_5$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+4\lambda_{HS_2}^{(1)}\lambda_5$	—	—

Terms 4 of 6

Column	Representative term structure
T1	$-9g_2^2\lambda_5$
T2	$+2\left(\lambda_7\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3	$+2\left(\lambda_8\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_5$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_5$	$+2\left(\lambda_7\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+2\left(\lambda_8\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_5$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_5$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_5$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_5$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_5$	$+3\left(\lambda_7\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+3\left(\lambda_8\right)^*\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_5$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_5$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_5$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$
T2	$+4\lambda_{11} (\lambda_7)^*$
T3	$+4(\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$	$+4\lambda_{11} (\lambda_7)^*$	$+4(\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (\lambda_7)^*$	$+4\lambda_{11} (\lambda_7)^*$	$+4(\lambda_8)^* ((\lambda_{10})^* - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_5

Terms 6 of 6

Column	Representative term structure
T1	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr} (Y_d \ (Y_d)^\dagger) + 2 \text{Tr} (Y_e \ (Y_e)^\dagger) + 6 \text{Tr} (Y_u \ (Y_u)^\dagger) + \frac{\lambda_{12}^{(A)}}{6} + \frac{2\lambda_{HS1}^{(A)}}{3} + 2\lambda_{HS2}^{(A)} \right)$
T2	$\lambda_5 (T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr} (Y_d \ (Y_d)^\dagger) + 2 \text{Tr} (Y_e \ (Y_e)^\dagger) + 6 \text{Tr} (Y_u \ (Y_u)^\dagger) - 4\lambda_{HS1}^{(A)})$
T3	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr} (Y_d \ (Y_d)^\dagger) + 2 \text{Tr} (Y_e \ (Y_e)^\dagger) + 6 \text{Tr} (Y_u \ (Y_u)^\dagger) - \frac{4\lambda_{12}^{(A)}}{3} - 2\lambda_{12}^{(\times)} - 2\lambda_{HS1}^{(A)} - \frac{4\lambda_{HS2}^{(A)}}{3} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T_1	T_2	T_3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + \frac{\lambda_1^{(A)}}{6} + \frac{2\lambda_1^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + \frac{\lambda_1^{(A)}}{6} + \frac{2\lambda_1^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + \frac{\lambda_1^{(A)}}{6} + \frac{2\lambda_1^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + \frac{\lambda_1^{(A)}}{6} + \frac{2\lambda_1^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\lambda_5 \left(\frac{3}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + \frac{\lambda_1^{(A)}}{6} + \frac{2\lambda_1^{(A)}}{3} + 2\lambda_{HS_2}^{(A)} \right)$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - 4\lambda_{HS_1}^{(A)})$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - 4\lambda_{HS_1}^{(A)})$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - 4\lambda_{HS_1}^{(A)})$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - 4\lambda_{HS_1}^{(A)})$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$\lambda_5(T_\nu^{(1)} + 2T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - 4\lambda_{HS_1}^{(A)})$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - \frac{4\lambda_1^{(A)}}{3} - 2\lambda_{12}^{(A)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_1^{(A)}}{3} \right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - \frac{4\lambda_1^{(A)}}{3} - 2\lambda_{12}^{(A)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_1^{(A)}}{3} \right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - \frac{4\lambda_1^{(A)}}{3} - 2\lambda_{12}^{(A)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_1^{(A)}}{3} \right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - \frac{4\lambda_1^{(A)}}{3} - 2\lambda_{12}^{(A)} - 2\lambda_{HS_1}^{(A)} - \frac{4\lambda_1^{(A)}}{3} \right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$\lambda_5 \left(T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d(Y_d)^\dagger) + 2 \text{Tr}(Y_\nu(Y_\nu)^\dagger) + 6 \text{Tr}(Y_\nu(Y_\nu)^\dagger) - \frac{4\lambda_1^{(A)}}{3} - 2\lambda_{12}^{(A)} - 2\lambda_{HS_1}^{(A)} - \frac{4\$

$\beta_{\lambda_{12}^{(A)}}^{(1)}$

Quantity: $\lambda_{12}^{(A)}$

$$16\pi^2\,\mu\,\frac{d\lambda_{12}^{(A)}}{d\mu}=\beta_{\lambda_{12}^{(A)}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$
T2	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$
T3	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{12}^{(A)}(2T_\nu^{(1)}+2T_\nu^{(2)})$	$-\frac{8}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$+\frac{4}{3}\lambda_5\,(\lambda_5)^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\frac{2}{3}\text{Tr}\big(y_1\,(y_1)^\dagger\,y_2\,(y_2)^\dagger\big)$	$-\frac{2}{9}\lambda_5\,(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{2}{3} \text{Tr} \left(y_1 \ (y_1)^\dagger \ y_2 \ (y_2)^\dagger \right)$	$-\frac{2}{9} \lambda_5 \ (\lambda_5)^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{2}{3} \text{Tr} \left(y_1 \ (y_1)^\dagger \ y_2 \ (y_2)^\dagger \right)$	$-\frac{2}{9} \lambda_5 \ (\lambda_5)^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+\frac{2}{3} \text{Tr} \left(y_1 \ (y_1)^\dagger \ y_2 \ (y_2)^\dagger \right)$	$-\frac{2}{9} \lambda_5 \ (\lambda_5)^*$

Quantity: $\lambda_{12}^{(A)}$

Terms 2 of 8

Column	Representative term structure
T1	$-9g_2^2\lambda_{12}^{(A)}$
T2	$-6g_1^2\lambda_{12}^{(A)}$
T3	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-9g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-24g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-24g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-24g_2^2\lambda_{12}^{(A)}$	$-6g_1^2\lambda_{12}^{(A)}$	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-24g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-24g_2^2\lambda_{12}^{(A)}$	—	$+8\lambda_{12}^{(1)}\lambda_{12}^{(A)}$

Quantity: $\lambda_{12}^{(A)}$

Terms 3 of 8

Column	Representative term structure
T1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$
T2	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	$+\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_{12}^{(A)}\lambda_{S_1}^{(1)}$	$+2\lambda_{12}^{(A)}\lambda_{S_2}^{(1)}$	—

Terms 4 of 8

Column	Representative term structure
T1	$+ \lambda_{10} \left(\frac{2}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + \frac{4}{3} \left(5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^* \right) \right)$
T2	$+ \lambda_{11} \left(\frac{2}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + \frac{4}{3} \left(\left(\lambda_{10} \right)^* + 5 \left(\lambda_{11} \right)^* \right) \right)$
T3	$+ \frac{2}{3} \left(\left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^* \right) \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_{10} \left(\frac{2}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + \frac{4}{3} \left(5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^* \right) \right) + \lambda_{11} \left(\frac{2}{3} \text{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + \frac{4}{3} \left(\left(\lambda_{10} \right)^* + 5 \left(\lambda_{11} \right)^* \right) \right) + \frac{2}{3} \left(\left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^* \right) \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right)$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 5 of 8

Column	Representative term structure
T1	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$
T2	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$
T3	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+g_1^2\left(48g_2^2-30\lambda_{12}^{(A)}\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{64}{3}\lambda_9\left(\lambda_9\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+g_1^2\left(18g_2^2-15\lambda_{12}^{(A)}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+g_1^2\left(48g_2^2-30\lambda_{12}^{(A)}\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_1^2\left(-2g_2^2-3\lambda_{12}^{(A)}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+g_1^2\left(6g_2^2-15\lambda_{12}^{(A)}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+g_1^2\left(16g_2^2-30\lambda_{12}^{(A)}\right)$

Quantity: $\lambda_{12}^{(A)}$

Terms 6 of 8

Column	Representative term structure
T1	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$
T2	$+ \lambda_{10} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} (5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^*) \right)$
T3	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$	$+ \lambda_{10} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} (5 \left(\lambda_{10} \right)^* + \left(\lambda_{11} \right)^*) \right)$	$+ \lambda_{11} \left(\text{Tr} \left(y_1 \left(y_2 \right)^{\dagger} \right) + \frac{4}{3} ((\lambda_{10})^* + 5 \left(\lambda_{11} \right)^*) \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$\lambda_{12}^{(A)}(3T_{\nu}^{(1)}+3T_{\nu}^{(2)})$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{12}^{(A)}$

Terms 7 of 8

Column	Representative term structure						
T1	$+\left((\lambda_{10})^*+(\lambda_{11})^*\right)\mathrm{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)$						
T2	$-\frac{8}{3}\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$						
T3	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$						

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\left((\lambda_{10})^*+(\lambda_{11})^*\right)\mathrm{Tr}\left(y_2\left(y_1\right)^{\dagger}\right)$	$-\frac{8}{3}\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+\frac{2}{3}\mathrm{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$\lambda_{12}^{(A)}\left(2T_{\nu}^{(1)}+2T_{\nu}^{(2)}+\frac{16\lambda_{12}^{(\times)}}{3}\right)$

Quantity: $\lambda_{12}^{(A)}$

Terms 8 of 8

Column	Representative term structure
T1	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$
T2	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$
T3	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+16\lambda_{12}^{(A)}\lambda_{S_1}^{(A)}$	$+16\lambda_{12}^{(A)}\lambda_{S_2}^{(A)}$	$+3\lambda_{HS_1}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Quantity: $\lambda_{12}^{(\times)}$

$$16\pi^2\,\mu\frac{d\lambda_{12}^{(\times)}}{d\mu}=\beta_{\lambda_{12}^{(\times)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	
T2	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	
T3	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	−2	1	−1	3	0	2	− $\frac{1}{2}$	—	—	—
T3-A	−1	1	− $\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	−2	2	−1	2	0	1	− $\frac{1}{2}$	—	—	—
T3-B	−1	2	− $\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	−2	2	−1	2	0	3	− $\frac{1}{2}$	—	—	—
T3-C	−1	2	− $\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	−2	3	−1	1	0	2	− $\frac{1}{2}$	—	—	—
T3-D	−1	3	− $\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	−2	3	−1	3	0	2	− $\frac{1}{2}$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$
T3-E	−1	3	− $\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{12}^{(1)}\left(\frac{1}{2}T_{\nu}^{(1)}+\frac{1}{2}T_{\nu}^{(2)}+8\lambda_{12}^{(\times)}+\lambda_{S_1}^{(A)}+\lambda_{S_2}^{(A)}\right)$	$+\lambda_{12}^{(\times)}(2T_{\nu}^{(1)}+2T_{\nu}^{(2)})$	$-\frac{7}{6}\text{Tr}\Big(y_1\ (y_1)^{\dagger}\ y_2\ (y_2)^{\dagger}\Big)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 2 of 6

Column	Representative term structure
T1	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$
T2	$+\frac{33g_2^4}{4}$
T3	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{1}{2}\lambda_5\left(\lambda_5\right)^*$	$+\frac{33g_2^4}{4}$	$+g_2^2\left(-6\lambda_{12}^{(1)}-24\lambda_{12}^{(\times)}\right)$

Quantity: $\lambda_{12}^{(\times)}$

Terms 3 of 6

Column	Representative term structure
T1	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{2} - 6\lambda_{12}^{(\times)} \right)$
T2	$+\lambda_{12}^{(1)2}$
T3	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{2} - 6\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{4} - 3\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2 \left(-\frac{3\lambda_{12}^{(1)}}{2} - 6\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2 \left(-\frac{15\lambda_{12}^{(1)}}{4} - 15\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2 \left(-\frac{15\lambda_{12}^{(1)}}{2} - 30\lambda_{12}^{(\times)} \right)$	$+\lambda_{12}^{(1)2}$	$+\lambda_{S_1}^{(1)}(2\lambda_{12}^{(1)} + 3\lambda_{12}^{(\times)})$

Quantity: $\lambda_{12}^{(\times)}$

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$
T2	$+4\lambda_{12}^{(A)2}$
T3	$+11\lambda_{12}^{(\times)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\lambda_{S_2}^{(1)}(2\lambda_{12}^{(1)}+3\lambda_{12}^{(\times)})$	$+4\lambda_{12}^{(A)2}$	$+11\lambda_{12}^{(\times)2}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 5 of 6

Column	Representative term structure
T1	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$
T2	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$
T3	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\lambda_{12}^{(\times)}\lambda_{S_1}^{(A)}$	$-\lambda_{12}^{(\times)}\lambda_{S_2}^{(A)}$	$+\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}$

Quantity: $\lambda_{12}^{(\times)}$

Terms 6 of 6

Column	Representative term structure
T1	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+\frac{3g_1^4}{16}$
T3	$+12g_1^4$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{7}{6}\text{Tr}\left(y_1\left(y_2\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+\frac{3g_1^4}{16}$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$+\frac{27g_1^4}{16}$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+12g_1^4$

$\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{HS_1}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_1}^{(A)}}{d\mu}=\beta_{\lambda_{HS_1}^{(A)}}^{(1)}$$

Terms 1 of 8

Column	Representative term structure
T1	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T2	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$\lambda_{HS_1}^{(A)}(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{8}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{16}{3}\lambda_5\ (\lambda_5)^*$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	$-\frac{4}{3}\text{Tr}\big(y_1\ (y_1)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{3}\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_5 (\lambda_5)^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_5 (\lambda_5)^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_5 (\lambda_5)^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_5 (\lambda_5)^*$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$-\frac{4}{3} \text{Tr} \left(y_1 (y_1)^\dagger Y_e (Y_e)^\dagger \right)$	$-\frac{4}{3} \lambda_5 (\lambda_5)^*$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 2 of 8

Column	Representative term structure
T1	$+g_1^2\left(-12g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$
T2	$-9g_2^2\lambda_{HS_1}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+g_1^2\left(-12g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+g_1^2\left(-12g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$-9g_2^2\lambda_{HS_1}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+g_1^2\left(-4g_2^2-\frac{15\lambda_{HS_1}^{(A)}}{2}\right)$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 3 of 8

Column	Representative term structure
T1	$+2\lambda_1\lambda_{HS_1}^{(A)}$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$
T3	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+2\lambda_1\lambda_{HS_1}^{(A)}$	$+8\lambda_{HS_1}^{(1)}\lambda_{HS_1}^{(A)}$	$+2\lambda_{HS_1}^{(A)}\lambda_{S_1}^{(1)}$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 4 of 8

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$
T2	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(-\frac{2}{3}\text{Tr}\left(y_1\left(y_2\right)^\dagger\right)-\frac{4}{3}\left(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*+(\lambda_{11})^*\right)\right)$	$-\frac{2}{3}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+\frac{64}{3}\lambda_7\left(\lambda_7\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 5 of 8

Column	Representative term structure
T1	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T2	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$
T3	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{4}{3}\lambda_{11}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$	$+g_1^2\left(-6g_2^2-3\lambda_{HS_1}^{(A)}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	$+g_1^2\left(-2g_2^2-3\lambda_{HS_1}^{(A)}\right)$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	$+g_1^2\left(-2g_2^2-3\lambda_{HS_1}^{(A)}\right)$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$-\frac{3g_1^2\lambda_{HS_1}^{(A)}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 6 of 8

Column	Representative term structure
T1	$+g_1^2 \left(6g_2^2 - 3\lambda_{HS_1}^{(A)} \right)$
T2	$+g_1^2 \left(12g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right)$
T3	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+g_1^2 \left(6g_2^2 - 3\lambda_{HS_1}^{(A)} \right)$	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_1^2 \left(12g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right)$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+g_1^2 \left(6g_2^2 - 3\lambda_{HS_1}^{(A)} \right)$	—	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_1^2 \left(12g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right)$	—	$\lambda_{HS_1}^{(A)} (3T_\nu^{(1)} + 6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right) + 2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right) + 6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_1^2 \left(2g_2^2 - 3\lambda_{HS_1}^{(A)} \right)$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+g_1^2 \left(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right)$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2 \left(2g_2^2 - 3\lambda_{HS_1}^{(A)} \right)$	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2 \left(4g_2^2 - \frac{15\lambda_{HS_1}^{(A)}}{2} \right)$	—	—

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 7 of 8

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr}(y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + (\lambda_{11})^*) \right)$
T2	$-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$
T3	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \left(-\text{Tr}(y_1 (y_2)^\dagger) - \frac{4}{3} (2 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* + (\lambda_{11})^*) \right) - \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^* \text{Tr}(y_2 (y_1)^\dagger)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$\lambda_{HS_1}^{(A)} (2T_\nu^{(1)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger) + 16 \lambda_{S_1}^{(A)})$

Quantity: $\lambda_{HS_1}^{(A)}$

Terms 8 of 8

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$
T2	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_1}^{(A)}}{2}$	$+12\lambda_{12}^{(A)}\lambda_{HS_2}^{(A)}$

$$\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Quantity: $\lambda_{HS_2}^{(A)}$

$$16\pi^2\mu\frac{d\lambda_{HS_2}^{(A)}}{d\mu}=\beta_{\lambda_{HS_2}^{(A)}}^{(1)}$$

Terms 1 of 7

Column	Representative term structure
T1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$
T2	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$
T3	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$+\frac{8}{3}\lambda_5\ (\lambda_5)^*$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\ (\lambda_5)^*$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\ (\lambda_5)^*$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\ (\lambda_5)^*$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big)+16\lambda_{S_2}^{(A)})$	$+\frac{4}{3}\text{Tr}\big(y_2\ (y_2)^\dagger\ Y_e\ (Y_e)^\dagger\big)$	$-\frac{8}{9}\lambda_5\ (\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)+16\lambda_{S_2}^{(A)})+\frac{4}{3}\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$	$-\frac{8}{9}\lambda_5\left(\lambda_5\right)^*$	

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$
T2	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$
T3	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	$-\frac{3g_1^2\lambda_{HS_2}^{(A)}}{2}$	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{33g_2^2\lambda_{HS_2}^{(A)}}{2}$	—	$+2\lambda_1\lambda_{HS_2}^{(A)}$

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$
T2	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$
T3	$+g_1^2\left(2g_2^2-3\lambda_{HS_2}^{(A)}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(2g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(6g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(6g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(18g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(2g_2^2-3\lambda_{HS_2}^{(A)}\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	$+g_1^2\left(6g_2^2-15\lambda_{HS_2}^{(A)}\right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)}$	$+2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}$	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$
T2	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(4g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+g_1^2\left(8g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+g_1^2\left(24g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	$\lambda_{HS_2}^{(A)}(2T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$	$+8\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+g_1^2\left(12g_2^2-\frac{15\lambda_{HS_2}^{(A)}}{2}\right)$	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+g_1^2\left(24g_2^2-\frac{51\lambda_{HS_2}^{(A)}}{2}\right)$	—	$-4\text{Tr}\left(y_2\left(y_2\right)^\dagger Y_e\left(Y_e\right)^\dagger\right)$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2 \left(4g_2^2 - \frac{15\lambda_{HS_2}^{(A)}}{2} \right)$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+g_1^2 \left(8g_2^2 - \frac{51\lambda_{HS_2}^{(A)}}{2} \right)$	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 5 of 7

Column	Representative term structure
T1	$+16\lambda_5\left(\lambda_5\right)^*$
T2	$-9g_2^2\lambda_{HS_2}^{(A)}$
T3	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+16\lambda_5\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{HS_2}^{(A)}$	$+\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{10})^*))$
T2	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2 (y_1)^\dagger)$
T3	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-2\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{10})^*))$	$-2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2 (y_1)^\dagger)$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(-3\text{Tr}(y_1 (y_2)^\dagger) - 4(2\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^* + (\lambda_{10})^*))$	$-3\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\text{Tr}(y_2 (y_1)^\dagger)$	$-4\lambda_{10}\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: $\lambda_{HS_2}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$+64\lambda_8\left(\lambda_8\right)^*$
T2	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+64\lambda_8\left(\lambda_8\right)^*$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+64\lambda_8\left(\lambda_8\right)^*$	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	$\lambda_{HS_2}^{(A)}\left(3T_\nu^{(2)}+6\operatorname{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\operatorname{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\operatorname{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	$+12\lambda_{12}^{(A)}\lambda_{HS_1}^{(A)}$

$$\beta_{\lambda_7}^{(1)}$$

Quantity: λ_7

$$16\pi^2\,\mu\,\frac{d\lambda_7}{d\mu}=\beta_{\lambda_7}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_7(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+(\lambda_5)^*\text{Tr}\big(y_2\,(y_1)^\dagger\big)$
T3	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_7(2T_\nu^{(1)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))+(\lambda_5)^*\text{Tr}\big(y_2\,(y_1)^\dagger\big)$	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-2\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}(\lambda_5)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+2\lambda_{11}(\lambda_5)^*$
T2	$-9g_2^2\lambda_7$
T3	$-3g_1^2\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{11}(\lambda_5)^*$	$-9g_2^2\lambda_7$	$-3g_1^2\lambda_7$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_7

Terms 3 of 5

Column	Representative term structure
T1	$+2\lambda_1\lambda_7$
T2	$+8\lambda_{HS_1}^{(1)}\lambda_7$
T3	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_7$	$+8\lambda_{HS_1}^{(1)}\lambda_7$	$+\frac{4\lambda_{HS_1}^{(A)}}{3}\lambda_7$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_7\lambda_{S_1}^{(1)}$
T2	$+4\lambda_8\lambda_9$
T3	$\lambda_7(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_7\lambda_{S_1}^{(1)}$	$+4\lambda_8\lambda_9$	$\lambda_7(3T_\nu^{(1)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—

$$\beta_{\lambda_{12}}^{(1)}$$

Quantity: λ_{12}

$$16\pi^2\,\mu\frac{d\lambda_{12}}{d\mu}=\beta_{\lambda_{12}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure	
T1	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)$	
T2	$+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)$	
T3	$+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{HS_1}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)+\lambda_{HS_1}^{(A)}\left(-\frac{1}{8}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)+\lambda_{HS_2}^{(1)}\left(\frac{1}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$\lambda_{HS_1}^{(1)}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{11}\right)+\lambda_{HS_1}^{(A)}\left(-\frac{3}{16}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)-\frac{\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}-\frac{\lambda_{11}}{4}\right)+\lambda_{HS_2}^{(1)}\left(\frac{3}{4}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+4\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}+3\lambda_{10}\right)$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda_{HS_2}^{(A)} \left(-\frac{3}{8} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{10}}{4} \right)$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$
T3	$-4 \text{Tr}(y_2 (y_1)^\dagger Y_e (Y_e)^\dagger)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HS_2}^{(A)} \left(-\frac{3}{8} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{10}}{4} \right)$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} (T_\nu^{(1)} + T_\nu^{(2)} + 6 \text{Tr}(Y_d (Y_d)^\dagger) + 2 \text{Tr}(Y_e (Y_e)^\dagger) + 6 \text{Tr}(Y_u (Y_u)^\dagger))$	$-4 \text{Tr}(y_2 (y_1)^\dagger Y_e (Y_e)^\dagger)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HS_2}^{(A)} \left(-\frac{9}{16} \text{Tr}(y_2 (y_1)^\dagger) - \frac{3\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2} - \frac{3\lambda_{10}}{4} \right)$	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+6\lambda_9 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$
T2	$-6(\lambda_8)^*(\lambda_5)^*$
T3	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+6\lambda_9 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6(\lambda_8)^*(\lambda_5)^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+6\lambda_9 \left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}} \right)^*$	$-6(\lambda_8)^*(\lambda_5)^*$	$-9g_2^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$-3g_1^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$+2\lambda_{12}^{(1)} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3	$+3\lambda_1 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_1^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+2\lambda_{12}^{(1)} \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$+3\lambda_1 \lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$-6\lambda_7\lambda_5$
T2	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-6\lambda_7\lambda_5$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-6\lambda_7\lambda_5$	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\left(\frac{3}{2}T_\nu^{(1)}+\frac{3}{2}T_\nu^{(2)}+6\text{Tr}\left(Y_d\left(Y_d\right)^\dagger\right)+2\text{Tr}\left(Y_e\left(Y_e\right)^\dagger\right)+6\text{Tr}\left(Y_u\left(Y_u\right)^\dagger\right)\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$16\pi^2\,\mu\,\frac{d\lambda_8}{d\mu}=\beta_{\lambda_8}^{(1)}$

Terms 1 of 4

Column	Representative term structure
T1	$\lambda_8(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$
T2	$+(\lambda_5)^*\text{Tr}\big(y_1\,(y_2)^\dagger\big)$
T3	$+4\lambda_7\,(\lambda_9)^*$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_8(2T_\nu^{(2)}+6\text{Tr}\big(Y_d\,(Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\,(Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\,(Y_u)^\dagger\big))$	$+(\lambda_5)^*\text{Tr}\big(y_1\,(y_2)^\dagger\big)$	$+4\lambda_7\,(\lambda_9)^*$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$+4\lambda_7\,(\lambda_9)^*$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_8

Terms 2 of 4

Column	Representative term structure	
T1	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	
T2	$-9g_2^2\lambda_8$	
T3	$-3g_1^2\lambda_8$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\left(\lambda_5\right)^*\left(\left(\lambda_{10}\right)^*-\left(\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}\right)^*\right)$	$-9g_2^2\lambda_8$	$-3g_1^2\lambda_8$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_8

Terms 3 of 4

Column	Representative term structure
T1	$+2\lambda_1\lambda_8$
T2	$+8\lambda_{HS_2}^{(1)}\lambda_8$
T3	$+4\lambda_{HS_2}^{(A)}\lambda_8$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_1\lambda_8$	$+8\lambda_{HS_2}^{(1)}\lambda_8$	$+4\lambda_{HS_2}^{(A)}\lambda_8$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 4

Column	Representative term structure
T1	$+2\lambda_8\lambda_{S_2}^{(1)}$
T2	$\lambda_8(3T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$
T3	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\big(y_1\ (y_2)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_8\lambda_{S_2}^{(1)}$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_8\lambda_{S_2}^{(1)}$	$\lambda_8(3T_\nu^{(2)}+6\text{Tr}\big(Y_d\ (Y_d)^\dagger\big)+2\text{Tr}\big(Y_e\ (Y_e)^\dagger\big)+6\text{Tr}\big(Y_u\ (Y_u)^\dagger\big))$	$+\frac{3}{2}(\lambda_5)^*\text{Tr}\big(y_1\ (y_2)^\dagger\big)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

$\beta_{\lambda_{S_1}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_1}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_1}^{(A)}}{d\mu}=\beta_{\lambda_{S_1}^{(A)}}^{(1)}$$

Terms 1 of 6

Column	Representative term structure	
T1	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	
T2	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	
T3	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_1}^{(1)}\left(\frac{1}{2}T_\nu^{(1)}+13\lambda_{S_1}^{(A)}\right)$	$+4\lambda_{S_1}^{(A)}T_\nu^{(1)}$	$-\text{Tr}\Big(y_1\left(y_1\right)^\dagger y_1\left(y_1\right)^\dagger\Big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_1}^{(1)} \left(\frac{1}{2} T_\nu^{(1)} + 13 \lambda_{S_1}^{(A)} \right)$	$+4 \lambda_{S_1}^{(A)} T_\nu^{(1)}$	$-\text{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_1}^{(1)} \left(\frac{1}{2} T_\nu^{(1)} + 13 \lambda_{S_1}^{(A)} \right)$	$+4 \lambda_{S_1}^{(A)} T_\nu^{(1)}$	$-\text{Tr} \left(y_1 (y_1)^\dagger y_1 (y_1)^\dagger \right)$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 2 of 6

Column	Representative term structure	
T1	$+ \frac{3g_2^4}{4}$	
T2	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	
T3	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	$+g_1^2 \left(9g_2^2 - \frac{3\lambda_{S_1}^{(1)}}{2} - 12\lambda_{S_1}^{(A)} \right)$	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+ \frac{3g_2^4}{4}$	—	$+g_2^2 (-3\lambda_{S_1}^{(1)} - 24\lambda_{S_1}^{(A)})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_2^4}{4}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_1}^{(1)}}{2}-12\lambda_{S_1}^{(A)}\right)$	$+g_2^2(-3\lambda_{S_1}^{(1)}-24\lambda_{S_1}^{(A)})$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 3 of 6

Column	Representative term structure
T1	$+ \frac{3g_1^4}{2}$
T2	$+ \frac{\lambda_{12}^{(1)2}}{4}$
T3	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+ \frac{3g_1^4}{2}$	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+ \frac{3g_1^4}{32}$	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+ \frac{3g_1^4}{32}$	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+ \frac{3g_1^4}{2}$	$+ \frac{\lambda_{12}^{(1)2}}{4}$	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+ \frac{3g_1^4}{2}$	—	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+ \frac{3g_1^4}{32}$	—	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	$+ \frac{\lambda_{HS_1}^{(1)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3g_1^4}{32}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_1^4}{2}$	—	$+\frac{\lambda_{HS_1}^{(1)2}}{2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 4 of 6

Column	Representative term structure
T1	$+3\lambda_{HS_1}^{(A)2}$
T2	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$
T3	$+22\lambda_{S_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+3\lambda_{HS_1}^{(A)2}$	$+\frac{7\lambda_{S_1}^{(1)2}}{4}$	$+22\lambda_{S_1}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+3\lambda_{HS_1}^{(A)^2}$	$+ \frac{7\lambda_{S_1}^{(1)^2}}{4}$	$+22\lambda_{S_1}^{(A)^2}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 5 of 6

Column	Representative term structure	
T1	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}^{(1)}}{8}-3\lambda_{S_1}^{(A)}\right)$	
T2	$+\frac{3\lambda_{12}^{(1)2}}{4}$	
T3	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}^{(1)}}{8}-3\lambda_{S_1}^{(A)}\right)$	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}^{(1)}}{8}-3\lambda_{S_1}^{(A)}\right)$	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}^{(1)}}{8}-3\lambda_{S_1}^{(A)}\right)$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_1}^{(1)}}{8}-3\lambda_{S_1}^{(A)}\right)$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$

Quantity: $\lambda_{S_1}^{(A)}$

Terms 6 of 6

Column	Representative term structure
T1	$+12\lambda_{12}^{(A)2}$
T2	$-\frac{\lambda_{12}^{(\times)2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+12\lambda_{12}^{(A)2}$	$-\frac{\lambda_{12}^{(\times)2}}{2}$

$\beta_{\lambda_{11}}^{(1)}$
Quantity: λ_{11}

$$16\pi^2\,\mu\frac{d\lambda_{11}}{d\mu}=\beta_{\lambda_{11}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$
T2	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$
T3	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$	$+\lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	$+\lambda_{S_1}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+12\lambda_{11})$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+\lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{\lambda_{10}}{6}+\frac{5\lambda_{11}}{6}\right)$	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure									
T1	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$									
T2	$+ \lambda_{11} (3 T_\nu^{(1)} + T_\nu^{(2)})$									
T3	$- 4 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$									

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+ \lambda_9 (2 \operatorname{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$	$+ \lambda_{11} (3 T_\nu^{(1)} + T_\nu^{(2)})$	$- 4 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_9 (3 \operatorname{Tr} \left(y_1 \left(y_2 \right)^\dagger \right) + 4 ((\lambda_{10})^* + 5 (\lambda_{11})^*))$	—	$- 5 \operatorname{Tr} \left(y_1 \left(y_1 \right)^\dagger y_2 \left(y_1 \right)^\dagger \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure	
T1	$-9g_2^2\lambda_{11}$	
T2	$-3g_1^2\lambda_{11}$	
T3	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-9g_2^2\lambda_{11}$	$-3g_1^2\lambda_{11}$	$+2\lambda_{HS_1}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_{11}

Terms 4 of 5

Column	Representative term structure
T1	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$
T2	$+4\lambda_7\lambda_5$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{\lambda_{HS_1}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{6}$	$+4\lambda_7\lambda_5$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_1}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{11} \right)$
T2	$+ \lambda_{11} \left(\frac{9}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_1}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{11} \right)$	$+ \lambda_{11} \left(\frac{9}{2} T_\nu^{(1)} + \frac{3}{2} T_\nu^{(2)} \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$16\pi^2\,\mu\,\frac{d\lambda_9}{d\mu}=\beta_{\lambda_9}^{(1)}$$

Terms 1 of 6

Column	Representative term structure
T1	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$
T2	$+ \lambda_{10}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+2\lambda_{11})$
T3	$+ \lambda_{11}\ \text{Tr}\big(y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_9(2T_\nu^{(1)}+2T_\nu^{(2)})$	$+ \lambda_{10}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+2\lambda_{11})$	$+ \lambda_{11}\ \text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_9

Terms 2 of 6

Column	Representative term structure
T1	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$
T2	$+4\lambda_7\left(\lambda_8\right)^*$
T3	$-9g_2^2\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-2\text{Tr}\left(y_2\left(y_1\right)^{\dagger}y_2\left(y_1\right)^{\dagger}\right)$	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+4\lambda_7\left(\lambda_8\right)^*$	$-9g_2^2\lambda_9$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_9

Terms 3 of 6

Column	Representative term structure
T1	$-3g_1^2\lambda_9$
T2	$+8\lambda_{12}^{(1)}\lambda_9$
T3	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-3g_1^2\lambda_9$	$+8\lambda_{12}^{(1)}\lambda_9$	$+\frac{4\lambda_{12}^{(A)}}{3}\lambda_9$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Quantity: λ_9

Terms 4 of 6

Column	Representative term structure
T1	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$
T2	$+2\lambda_{S_1}^{(1)}\lambda_9$
T3	$+2\lambda_9\lambda_{S_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}{}^2$	$+2\lambda_{S_1}^{(1)}\lambda_9$	$+2\lambda_9\lambda_{S_2}^{(1)}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 6

Column	Representative term structure
T1	$+5\lambda_{10}^2$
T2	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$
T3	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+5\lambda_{10}^2$ $+5\lambda_{11}^2$	$\lambda_9(3T_\nu^{(1)}+3T_\nu^{(2)})$	$+\lambda_{10}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+2\lambda_{11}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 6 of 6

Column	Representative term structure
T1	$+\frac{3}{2}\lambda_{11}\text{Tr}\big(y_2\ (y_1)^\dagger\big)$
T2	$-\frac{5}{2}\text{Tr}\big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\big)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{3}{2}\lambda_{11}\text{Tr}\big(y_2\ (y_1)^\dagger\big)$	$-\frac{5}{2}\text{Tr}\big(y_2\ (y_1)^\dagger\ y_2\ (y_1)^\dagger\big)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$$16\pi^2\,\mu\frac{d\lambda_{10}}{d\mu}=\beta_{\lambda_{10}}^{(1)}$$

Terms 1 of 5

Column	Representative term structure
T1	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$
T2	$+ \lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$
T3	$+ \lambda_9(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$\lambda_{12}^{(1)}(\text{Tr}\big(y_2\ (y_1)^\dagger\big)+6\lambda_{10}+6\lambda_{11})$	$+ \lambda_{12}^{(A)}\left(\frac{1}{12}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+ \lambda_9(2\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	$+ \lambda_{12}^{(A)}\left(\frac{1}{8}\text{Tr}\big(y_2\ (y_1)^\dagger\big)+\frac{5\lambda_{10}}{6}+\frac{\lambda_{11}}{6}\right)$	$+ \lambda_9(3\text{Tr}\big(y_1\ (y_2)^\dagger\big)+4(5\ (\lambda_{10})^*+(\lambda_{11})^*))$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 2 of 5

Column	Representative term structure
T1	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10})$
T2	$+\lambda_{10}(T_\nu^{(1)}+3T_\nu^{(2)})$
T3	$-4\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\lambda_{S_2}^{(1)}(\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+12\lambda_{10})$	$+\lambda_{10}(T_\nu^{(1)}+3T_\nu^{(2)})$	$-4\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	$-5\text{Tr}\left(y_2\left(y_1\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 3 of 5

Column	Representative term structure
T1	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$
T2	$-9g_2^2\lambda_{10}$
T3	$-3g_1^2\lambda_{10}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+4\left(\lambda_8\right)^*\left(\lambda_5\right)^*$	$-9g_2^2\lambda_{10}$	$-3g_1^2\lambda_{10}$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 4 of 5

Column	Representative term structure
T1	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$
T2	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$
T3	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+2\lambda_{HS_2}^{(1)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}$	$-\frac{\lambda_{HS_2}^{(A)}\lambda_{HH^\dagger S_1^\dagger S_2^\dagger}^{\text{Cross}}}{2}$	$\lambda_{12}^{(1)}\left(\frac{3}{2}\text{Tr}\left(y_2\left(y_1\right)^\dagger\right)+6\lambda_{10}+6\lambda_{11}\right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—	—

Terms 5 of 5

Column	Representative term structure
T1	$+ \lambda_{S_2}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{10} \right)$
T2	$+ \lambda_{10} \left(\frac{3}{2} T_\nu^{(1)} + \frac{9}{2} T_\nu^{(2)} \right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+ \lambda_{S_2}^{(1)} \left(\frac{3}{2} \text{Tr} \left(y_2 \left(y_1 \right)^\dagger \right) + 12 \lambda_{10} \right)$	$+ \lambda_{10} \left(\frac{3}{2} T_\nu^{(1)} + \frac{9}{2} T_\nu^{(2)} \right)$
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	—

$\beta_{\lambda_{S_2}^{(A)}}^{(1)}$

Quantity: $\lambda_{S_2}^{(A)}$

$$16\pi^2\,\mu\frac{d\lambda_{S_2}^{(A)}}{d\mu}=\beta_{\lambda_{S_2}^{(A)}}^{(1)}$$

Terms 1 of 7

Column	Representative term	structure
T1	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	
T2	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	
T3	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\lambda_{S_2}^{(1)}\left(\frac{1}{2}T_\nu^{(2)}+13\lambda_{S_2}^{(A)}\right)$	$+4\lambda_{S_2}^{(A)}T_\nu^{(2)}$	$-\text{Tr}\left(y_2\left(y_2\right)^\dagger y_2\left(y_2\right)^\dagger\right)$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$\lambda_{S_2}^{(1)} \left(\frac{1}{2} T_\nu^{(2)} + 13 \lambda_{S_2}^{(A)} \right)$	$+4 \lambda_{S_2}^{(A)} T_\nu^{(2)}$	$-\text{Tr} \left(y_2 \ (y_2)^\dagger \ y_2 \ (y_2)^\dagger \right)$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$\lambda_{S_2}^{(1)} \left(\frac{1}{2} T_\nu^{(2)} + 13 \lambda_{S_2}^{(A)} \right)$	$+4 \lambda_{S_2}^{(A)} T_\nu^{(2)}$	$-\text{Tr} \left(y_2 \ (y_2)^\dagger \ y_2 \ (y_2)^\dagger \right)$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 2 of 7

Column	Representative term structure
T1	$+\frac{3g_2^4}{4}$
T2	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$
T3	$+\frac{\lambda_{12}^{(1)2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	$+\frac{\lambda_{12}^{(1)2}}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3g_2^4}{4}$	$+g_2^2(-3\lambda_{S_2}^{(1)}-24\lambda_{S_2}^{(A)})$	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 3 of 7

Column	Representative term structure
T1	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$
T2	$+3\lambda_{HS_2}^{(A) \ 2}$
T3	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{\lambda_{HS_2}^{(1) \ 2}}{2}$	$+3\lambda_{HS_2}^{(A) \ 2}$	$+\frac{7\lambda_{S_2}^{(1) 2}}{4}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$	$+3\lambda_{HS_2}^{(A)2}$	$+\frac{7\lambda_{S_2}^{(1)2}}{4}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{\lambda_{HS_2}^{(1)2}}{2}$	$+3\lambda_{HS_2}^{(A)2}$	$+\frac{7\lambda_{S_2}^{(1)2}}{4}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 4 of 7

Column	Representative term structure
T1	$+22\lambda_{S_2}^{(A)2}$
T2	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}^{(1)}}{8}-3\lambda_{S_2}^{(A)}\right)$
T3	$+\frac{3g_1^4}{32}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+22\lambda_{S_2}^{(A)2}$	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+22\lambda_{S_2}^{(A)2}$	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}^{(1)}}{8}-3\lambda_{S_2}^{(A)}\right)$	$+\frac{3g_1^4}{32}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{(A)2}$	—	$+\frac{3g_1^4}{2}$
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+22\lambda_{S_2}^{(A)2}$	$+g_1^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}^{(1)}}{8}-27\lambda_{S_2}^{(A)}\right)$	$+\frac{243g_1^4}{32}$
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+22\lambda_{S_2}^{(A)2}$	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+22\lambda_{S_2}^{(A)2}$	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+22\lambda_{S_2}^{(A)2}$	$+g_1^2\left(\frac{9g_2^2}{4}-\frac{3\lambda_{S_2}^{(1)}}{8}-3\lambda_{S_2}^{(A)}\right)$	$+\frac{3g_1^4}{32}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+22\lambda_{S_2}^{(A)2}$	—	$+\frac{3g_1^4}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+22\lambda_{S_2}^{(A)2}$	$+g_1^2\left(\frac{81g_2^2}{4}-\frac{27\lambda_{S_2}^{(1)}}{8}-27\lambda_{S_2}^{(A)}\right)$	$+\frac{243g_1^4}{32}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+22\lambda_{S_2}^{(A)^2}$	—	—

Quantity: $\lambda_{S_2}^{(A)}$

Terms 5 of 7

Column	Representative term structure	
T1	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	
T2	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$	
T3	$+24g_1^4$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$	$+24g_1^4$
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+g_1^2\left(9g_2^2-\frac{3\lambda_{S_2}^{(1)}}{2}-12\lambda_{S_2}^{(A)}\right)$	—	—
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	—	$+g_1^2\left(36g_2^2-6\lambda_{S_2}^{(1)}-48\lambda_{S_2}^{(A)}\right)$	$+24g_1^4$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 6 of 7

Column	Representative term structure
T1	$+\frac{3\lambda_{12}^{(1)2}}{4}$
T2	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$
T3	$+12\lambda_{12}^{(A)2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1	T2	T3
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—	—	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—	—	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—	—	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—	—	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—	—	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—	—	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—	—	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—	—	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—	—	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—	—	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—	—	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—	—	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—	—	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—	—	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—	—	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—	—	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—	—	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—	—	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—	—	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—	—	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{3\lambda_{12}^{(1)2}}{4}$	$+\lambda_{12}^{(1)}\lambda_{12}^{(\times)}$	$+12\lambda_{12}^{(A)2}$

Quantity: $\lambda_{S_2}^{(A)}$

Terms 7 of 7

Column	Representative term structure
T1	$-\frac{\lambda_{12}^{(\times)^2}}{2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	T1
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	—
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	—
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	—
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	—
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	—
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	—
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	—
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	—
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	—
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	—
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	—
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	—
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	—
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	—
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	—
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	—
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	—
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	—
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	—
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	—
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{\lambda_{12}^{(\times)^2}}{2}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{\lambda_{12}^{(\times)^2}}{2}$